PHYS 1210 Thermal Physics — Lecture 7: Ideal gases and the First Law

7.1 Ideal gas

An ideal gas obeys $PV = nRT$, with $R = 8.314\ \mathrm{J\,mol^{-1}K^{-1}}$. Its internal energy depends only on temperature: $U = nC_V T$.

7.2 The First Law

$\Delta U = Q - W$, where W is the work done *by* the gas, $W = \int_{V_1}^{V_2} P\,dV$.

- Isochoric (V const): $W = 0$, so $Q = nC_V\Delta T$.
- Isobaric (P const): $W = P\Delta V = nR\Delta T$, so $Q = nC_P\Delta T$ with $C_P = C_V + R$ (Mayer's relation).
- Isothermal (T const): $\Delta U = 0$, so $Q = W = nRT\ln(V_2/V_1)$.

The heat capacity ratio is $\gamma = C_P/C_V$. For a monatomic gas $C_V = \frac{3}{2}R$, so $\gamma = 5/3$.

7.3 Adiabatic processes

With $Q = 0$ and a reversible process, $PV^\gamma = \text{const}$. Combining with $PV = nRT$ gives

$$TV^\gamma = \text{const}.$$

The work done by the gas is $W = \dfrac{P_1V_1 - P_2V_2}{\gamma - 1}$.

7.4 Heat engines

Efficiency $\eta = W/Q_H = 1 - Q_C/Q_H$. A Carnot engine between reservoirs T_H and T_C has $\eta_C = 1 - T_C/T_H$.

Example: an engine between $T_H = 500\,\mathrm{K}$ and $T_C = 300\,\mathrm{K}$ has a maximum efficiency of $\eta_C = 1 - 300/500 = 60\%$.